camera. The recovery is faster and patient can be discharged on the next day.

- **Open cholecystectomy** is performed when laparoscopic cholecystectomy is not possible. Open surgery causes more pain and recovery time in hospital is around a week.
- Extra Corporeal Shock-Wave Lithotripsy (ECSW)

Supportive Management

- Intravenous fluids administration
- Nasogastric tube suction
- Pain management
- Antibiotics

Complication

- Cholangitis
- Necrosis
- Gallstoneileus leads to intestinal obstruction

2.13 Hernia

A hernia occurs when an organ pushes through an opening in the muscle or tissue that holds it in place.

Definition

Hernia is a localized bulge in the abdomen that occurs when there is a weakness in the muscular wall.

Causes

- Congenital - failure of the abdominal wall to close.
- Age – after 60 years
- Chronic cough
- Pregnancy - which puts pressure on the abdomen
- Constipation
- Lifting heavy weight
- Fluid in the abdomen, or ascites
- Abdominal surgery
- Obesity

Types

Inguinal hernia: Intestines push through the inguinal canal.

Incisional hernia: Resulting from an operated site.incision

Hiatal hernia: Part of the stomach protrudes up through the diaphragm into the chest cavity.

Umbilical hernia: A part of the bowel protrude through the belly button (umbilicus).

Congenital diaphragmatic hernias is a birth defect in which there is an abnormal opening in the diaphragm.

Sports hernia is a tear or strain of any tissue in the lower abdomen or groin due to gym activity.

Signs and Symptoms

- Swelling in the abdomen or in the groin which disappears when lying down.
- Pain on palpation
- Vomiting
- Feeling of weight in the abdomen.
- Constipation
- Discomfort in the abdomen or groin when coughing, lifting a weight or bending over.
- Fever
- Upper abdominal pain
- Chest pain

Diagnosis

- History collection
- Physical examination
- Abdominal X-rays
- Complete blood count, electrolytes.
- Ultrasonography of the abdomen
- CT abdomen

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